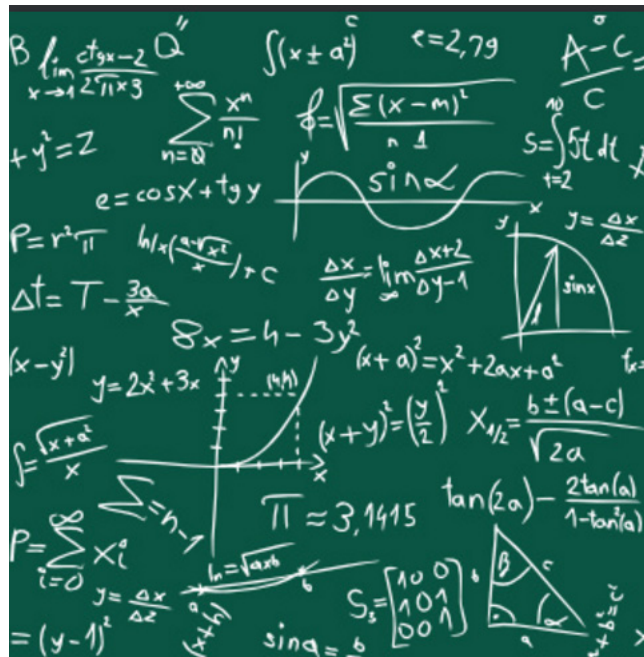


Limits of polynomials and rational functions



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- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

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1 Limits of polynomials at $\pm\infty$

- A polynomial of (maximum) degree n can be written as:

$$P_n(x) = \sum_{i=0}^n a_i x^i \quad \text{with } a_i \in \mathbb{R}, \text{ for } i \in \{0, 1, \dots, n\}, \text{ and } a_n \neq 0.$$

The value a_i is the **coefficient of the monomial** i , with $i = 0, 1, \dots, n$, while x^i is the **literal part** of the monomial. In this notation, without loss of generality, the monomials are ordered by increasing values of the exponents.¹

- Factoring out $a_n x^n$, that is, the last monomial, which is the one of maximum degree, we have:

$$P_n(x) = a_n x^n \left(\overbrace{\frac{a_0}{a_n x^n} + \frac{a_1}{a_n x^{n-1}} + \dots + \frac{a_{n-1}}{a_n x}}^{\rightarrow 0 \text{ as } x \rightarrow \pm\infty} + 1 \right)$$

$$\lim_{x \rightarrow \pm\infty} P_n(x) = \lim_{x \rightarrow \pm\infty} a_n x^n$$

To compute the limit of a polynomial as $x \rightarrow \pm\infty$, it is enough to compute the limit of the monomial of maximum degree.

Example 1: Limits of polynomials as $x \rightarrow \pm\infty$

For example:

$$\lim_{x \rightarrow +\infty} (8 - x^2 + 3x^3) = \lim_{x \rightarrow +\infty} 3x^3 \left(\underbrace{\frac{8}{3x^3} - \frac{1}{3x}}_{\rightarrow 0 \text{ as } x \rightarrow +\infty} + 1 \right) = \lim_{x \rightarrow +\infty} 3x^3 = +\infty$$

$$\lim_{x \rightarrow -\infty} (120 + 4x - 2x^3) = \lim_{x \rightarrow -\infty} -2x^3 \left(\underbrace{\frac{120}{-2x^3} + \frac{4}{-2x^2}}_{\rightarrow 0 \text{ as } x \rightarrow -\infty} + 1 \right) = \lim_{x \rightarrow -\infty} -2x^3 = +\infty$$

2 Limits of rational functions at $\pm\infty$

- Let f be a rational function (ratio of polynomials):

$$f(x) = \frac{P_n(x)}{P_m(x)}$$

¹If a monomial of degree i is missing we have $a_i = 0$; a_0 is the constant term since $x^0 = 1$.

where $P_n(x)$ and $P_m(x)$ are polynomials of degree n and m , respectively:

$$P_n(x) = \sum_{i=0}^n a_i x^i \quad \text{with } a_i \in \mathbb{R}, \text{ for } i = 0, 1, \dots, n, \text{ and } a_n \neq 0.$$

$$P_m(x) = \sum_{i=0}^m b_i x^i \quad \text{with } b_i \in \mathbb{R}, \text{ for } i = 0, 1, \dots, m, \text{ and } b_m \neq 0.$$

- Factoring out $a_n x^n$ in the numerator and $b_m x^m$ in the denominator, that is, the monomials of maximum degree, we have:

$$\frac{P_n(x)}{P_m(x)} = \frac{a_n x^n}{b_m x^m} \frac{\left\{ \overbrace{\frac{a_0}{a_n x^n} + \frac{a_1}{a_n x^{n-1}} + \dots + \frac{a_{n-1}}{a_n x} + 1}^{\rightarrow 0 \text{ as } x \rightarrow \pm\infty} \right\}}{\left\{ \underbrace{\frac{b_0}{b_m x^m} + \frac{b_1}{b_m x^{m-1}} + \dots + \frac{b_{m-1}}{b_m x} + 1}_{\rightarrow 0 \text{ as } x \rightarrow \pm\infty} \right\}}.$$

$$\lim_{x \rightarrow \pm\infty} \frac{P_n(x)}{P_m(x)} = \lim_{x \rightarrow \pm\infty} \frac{a_n x^n}{b_m x^m} = \lim_{x \rightarrow \pm\infty} \frac{a_n}{b_m} x^{n-m}$$

To compute the limit of a ratio of polynomials as $x \rightarrow \pm\infty$, it is enough to compute the limit of the ratio of the monomials of maximum degree.

Example 2: Limits of rational functions as $x \rightarrow \pm\infty$

For example:

$$\lim_{x \rightarrow -\infty} \frac{1 + 3x^3}{2 + x^2} = \lim_{x \rightarrow -\infty} \frac{3x^3}{x^2} = \lim_{x \rightarrow -\infty} 3x = -\infty$$

$$\lim_{x \rightarrow +\infty} \frac{1 + 3x^3}{1 - 10x + x^4} = \lim_{x \rightarrow +\infty} \frac{3x^3}{x^4} = \lim_{x \rightarrow +\infty} 3 \frac{1}{x} = 0$$

$$\lim_{x \rightarrow -\infty} \frac{7x^2 + 6x^5}{-4 + 7x^5} = \lim_{x \rightarrow -\infty} \frac{6x^5}{7x^5} = \lim_{x \rightarrow -\infty} \frac{6}{7} = \frac{6}{7}$$

To compute the limit of rational functions (without the constant term) as $x \rightarrow 0$ we must factor out, in the numerator and in the denominator, the powers of minimum degree.

Example 3: Limits of rational functions as $x \rightarrow 0$

For example:

$$\lim_{x \rightarrow 0} \frac{-2x^2 + 3x^3 - 6x^5}{x + x^2 - x^7} = \lim_{x \rightarrow 0} \frac{-2x^2 \left(\overbrace{1 - \frac{3}{2}x + 3x^3}^{\rightarrow 0 \text{ as } x \rightarrow 0} \right)}{x \left(\overbrace{1 + x - x^6}^{\rightarrow 0 \text{ as } x \rightarrow 0} \right)} = \lim_{x \rightarrow 0} -2x = 0$$

3 Limits of quotients of sums of powers with rational exponents

- The same rules hold for quotients of sums of powers with rational exponents:

as $x \rightarrow \infty$ we must factor out the power with the maximum exponent

as $x \rightarrow 0^+$ we must factor out the power with the minimum exponent

Raising to a power can also be defined with a negative base if the exponent is a rational number (fraction) with an odd denominator. Hence in this case we can compute the limit as $x \rightarrow 0$ (otherwise only as $x \rightarrow 0^+$).

Example 4: Limits of quotients of sums of powers with rational exponents

For example:

$$\lim_{x \rightarrow 0} \frac{x^{2/3}}{3x^{1/3} + x + x^2}$$

Factoring out in the denominator the power with the minimum exponent, we have:

$$\frac{x^{2/3}}{3x^{1/3} + x + x^2} = \frac{x^{2/3}}{3x^{1/3} \left(1 + \frac{x}{3x^{1/3}} + \frac{x^2}{3x^{1/3}} \right)} = \frac{x^{2/3}}{3x^{1/3} \left(1 + \frac{x^{2/3}}{3} + \frac{x^{5/3}}{3} \right)} = \frac{1}{3} x^{1/3} \frac{1}{\left(1 + \frac{x^{2/3}}{3} + \frac{x^{5/3}}{3} \right)}$$

hence

$$\lim_{x \rightarrow 0} \frac{x^{2/3}}{3x^{1/3} + x + x^2} = \frac{1}{3} \lim_{x \rightarrow 0} x^{1/3} \frac{1}{\left(1 + \underbrace{\frac{x^{2/3}}{3} + \frac{x^{5/3}}{3}}_{\rightarrow 0 \text{ as } x \rightarrow 0} \right)} = 0$$

Another option to compute limits as $x \rightarrow 0$ is to make a change of variable:

$$x = \frac{1}{y}, \text{ if } x \rightarrow 0^+ \text{ then } y \rightarrow +\infty, \text{ if } x \rightarrow 0^- \text{ then } y \rightarrow -\infty$$

and hence we reduce to the case of limits at $\pm\infty$.

Example 5: Computing the limit of rational functions as $x \rightarrow 0^+$ with a change of variable

We want to compute the following limit:

$$\lim_{x \rightarrow 0} \frac{x^{2/3}}{3x^{1/3} + x + x^2}, \quad f(x) = \frac{x^{2/3}}{3x^{1/3} + x + x^2}$$

Making the change of variable $x = \frac{1}{y}$, we therefore have to compute the following two limits:

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{y \rightarrow +\infty} \frac{\left(\frac{1}{y}\right)^{2/3}}{3\left(\frac{1}{y}\right)^{1/3} + \frac{1}{y} + \left(\frac{1}{y}\right)^2} \quad \text{and} \quad \lim_{x \rightarrow 0^-} f(x) = \lim_{y \rightarrow -\infty} \frac{\left(\frac{1}{y}\right)^{2/3}}{3\left(\frac{1}{y}\right)^{1/3} + \frac{1}{y} + \left(\frac{1}{y}\right)^2}$$

We have:

$$\frac{\left(\frac{1}{y}\right)^{2/3}}{3\left(\frac{1}{y}\right)^{1/3} + \frac{1}{y} + \left(\frac{1}{y}\right)^2} = \frac{y^{-2/3}}{3y^{-1/3} + y^{-1} + y^{-2}}$$

Factoring out in the denominator the power with the maximum exponent, we have:

$$\frac{y^{-2/3}}{3y^{-1/3} + y^{-1} + y^{-2}} = \frac{y^{-2/3}}{3y^{-1/3}\left(1 + \frac{y^{-1}}{3y^{-1/3}} + \frac{y^{-2}}{3y^{-1/3}}\right)} = \frac{1}{3} \frac{1}{y^{1/3}} \frac{1}{\left(1 + \frac{1}{3y^{2/3}} + \frac{1}{3y^{5/3}}\right)}$$

Hence:

$$\frac{1}{3} \lim_{y \rightarrow -\infty} \frac{1}{y^{1/3}} \frac{1}{\left(1 + \frac{1}{3y^{2/3}} + \frac{1}{3y^{5/3}}\right)} = \frac{1}{3} \lim_{y \rightarrow +\infty} \frac{1}{y^{1/3}} \frac{1}{\left(1 + \frac{1}{3y^{2/3}} + \frac{1}{3y^{5/3}}\right)} = 0$$

since, as $y \rightarrow \pm\infty$, we have

$$\frac{1}{y^{1/3}} \rightarrow 0 \quad \text{and} \quad \left(1 + \frac{1}{3y^{2/3}} + \frac{1}{3y^{5/3}}\right) \rightarrow 1 \quad \text{since} \quad \frac{1}{3y^{2/3}} \rightarrow 0, \quad \frac{1}{3y^{5/3}} \rightarrow 0.$$

Since the right limit and the left limit of $f(x)$ as $x \rightarrow 0$ exist and are equal to 0, then:

$$\lim_{x \rightarrow 0} f(x) = 0$$